

A STORYBOARD IN 9 FRAMES

Retrieval in plain terms

*Found it. Filed it.
Forgot it.*



Retrieval fetches material when a question comes in and puts it into the model's input. Finding it and handing it over are separate steps, and either can drop the evidence.

Six stages. Any one can lose it.

1

Source material

whatever holds the answer

:

2

Preparation and indexing

if it never got in, nothing finds it

:

3

Retrieval

candidates for this question

:

4

Ranking and filtering

most of them get dropped

:

5

Context assembly

what the model actually receives

:

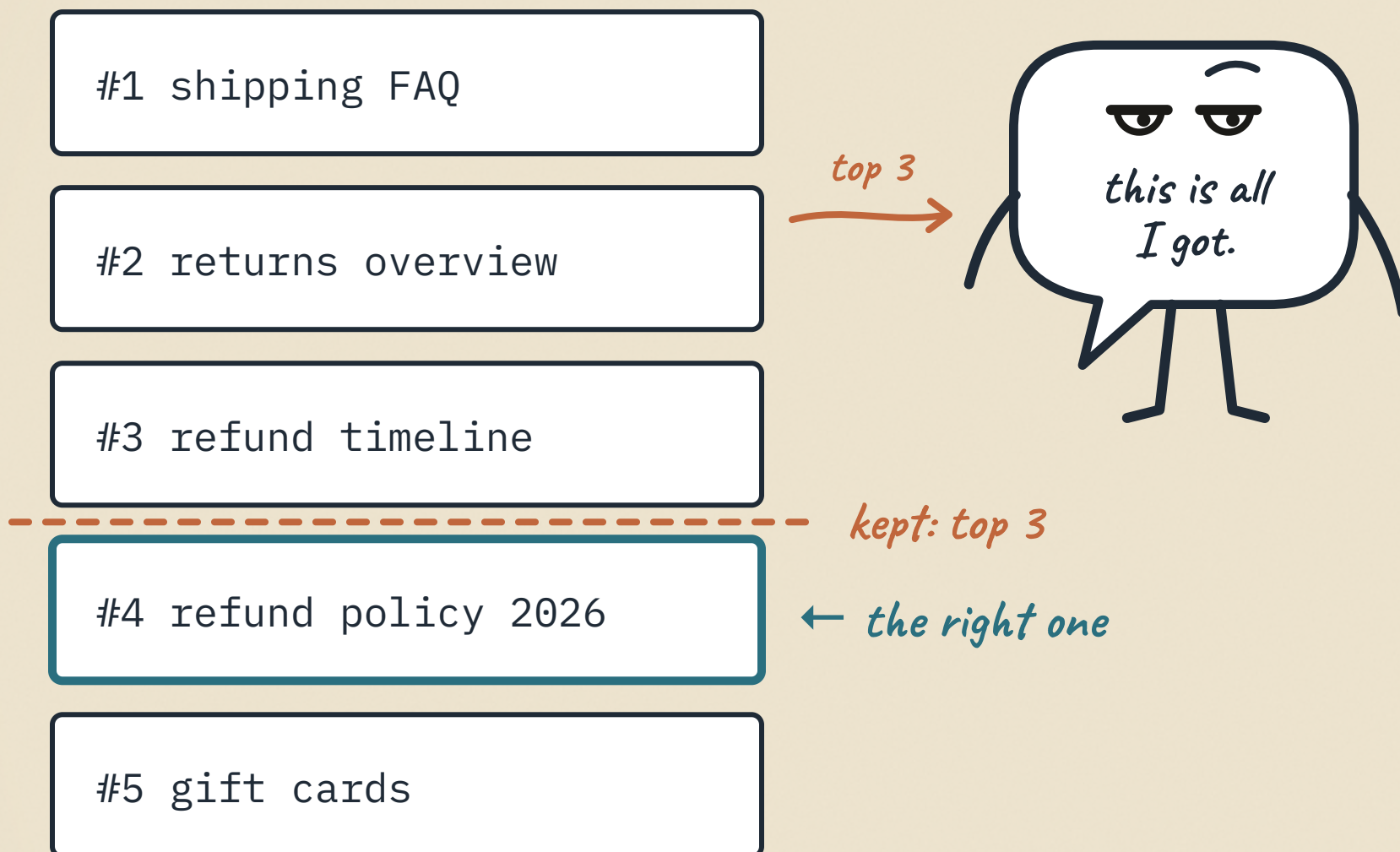
6

Generation

an answer from what it was handed

Small systems fold several of these into one search call. When the answer is wrong, usually exactly one of them lost the evidence.

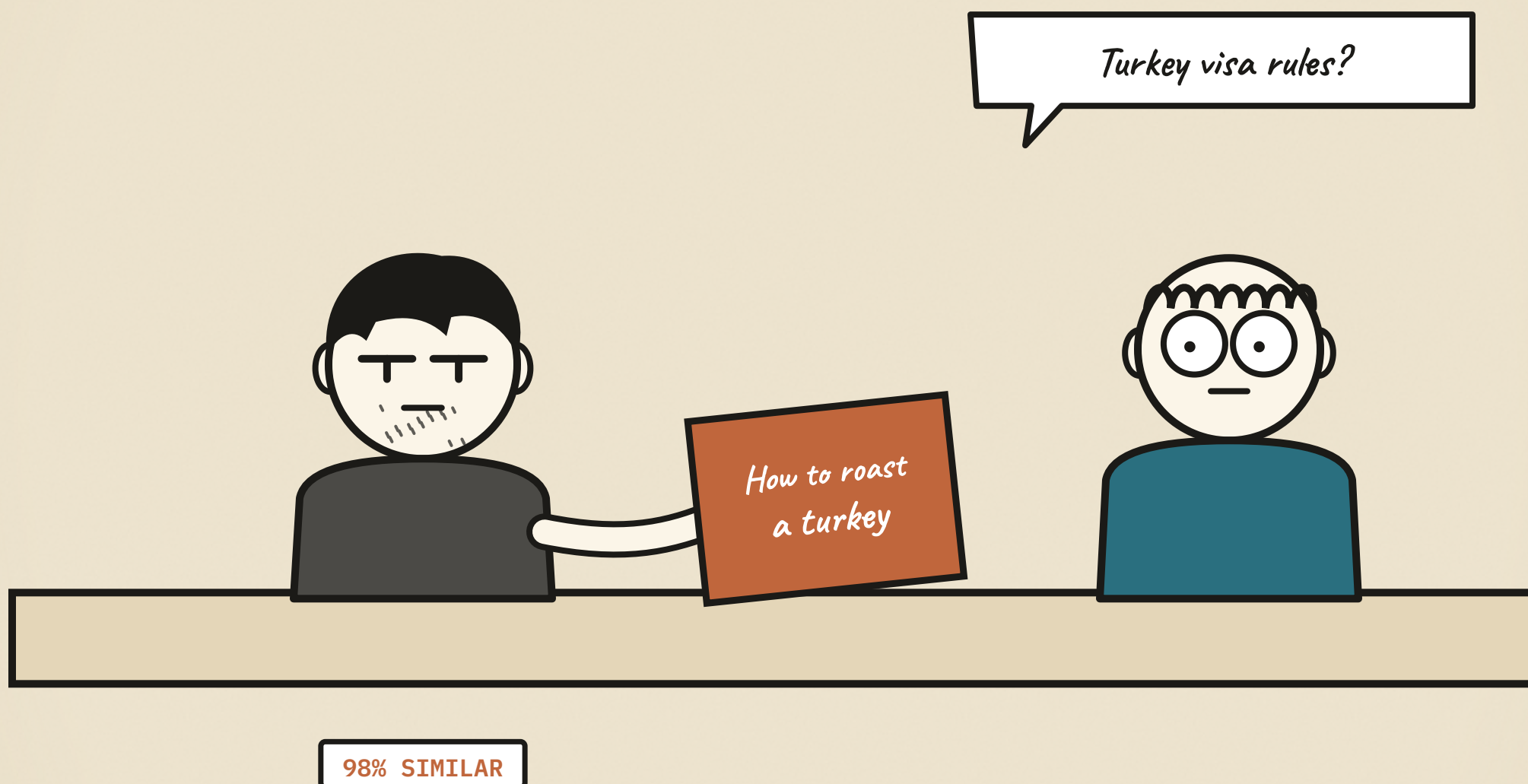
Found it. Dropped it.



retrieval succeeded.
the model never saw it.

Nothing in the search logs looks wrong, because nothing about the search was wrong. Log what was retrieved and what was assembled: they are different lists.

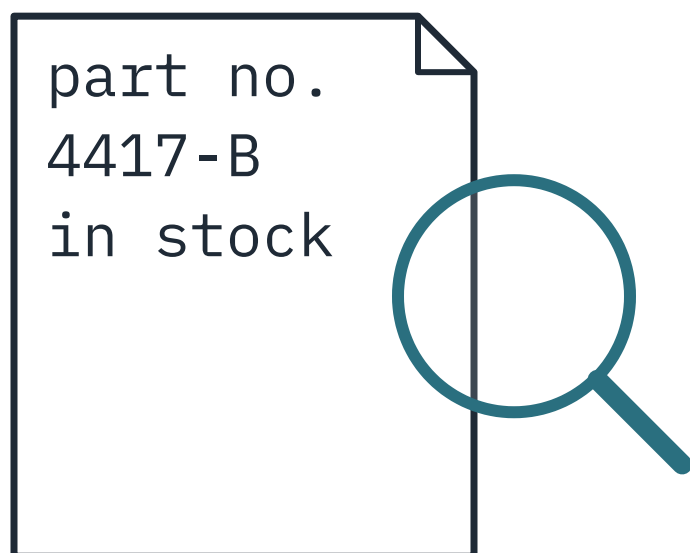
Similar isn't the answer.



Close in meaning. Wrong country.

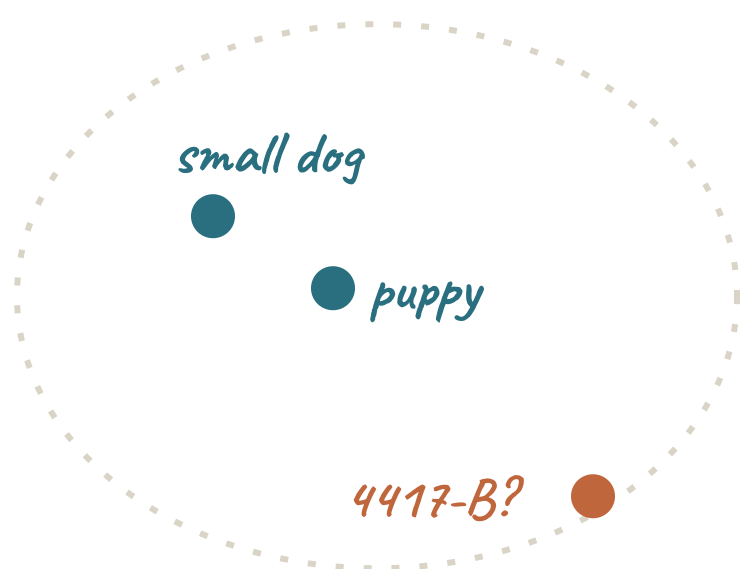
Two passages can be close in an embedding space and still answer different questions. Similar is not correct.

Words, or meaning?



Matching words

*finds exact IDs,
misses "puppy"
for "small dog"*



Matching meaning

*finds paraphrases,
misses exact IDs*

Lexical search misses a paraphrase; vector search misses an exact identifier. Hybrid systems run both, because the two fail in different places.

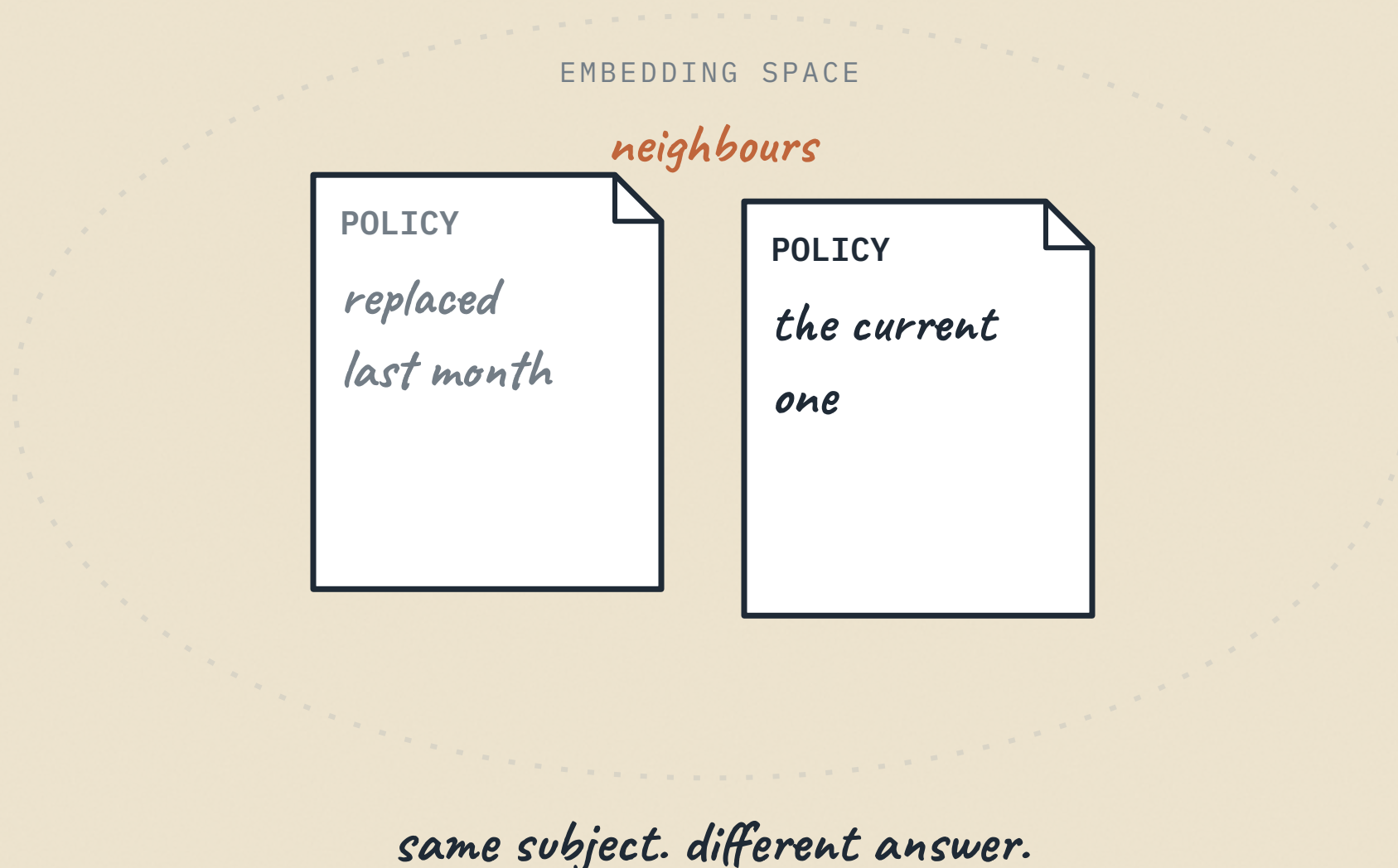
It didn't fix it. It footnoted it.



Wrong, now with a citation.

Retrieval does not guarantee the evidence is relevant, current or correct, or that the model uses it well. Given a wrong document, it is wrong with a source attached, which is harder to catch.

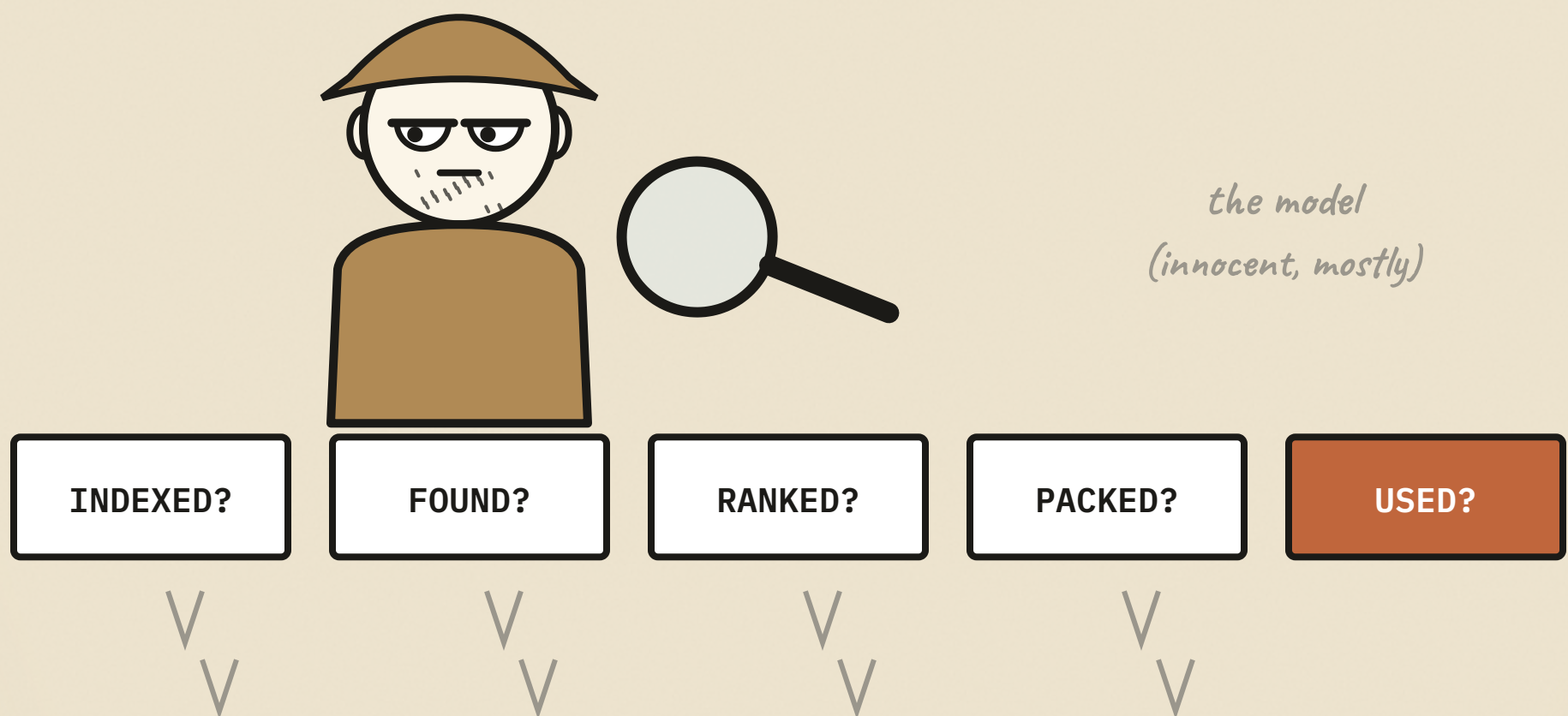
Close isn't correct.



~~nearest~~ → **still true?**

Two passages can sit close together in an embedding space and still answer different questions. That is retrieval working as designed.

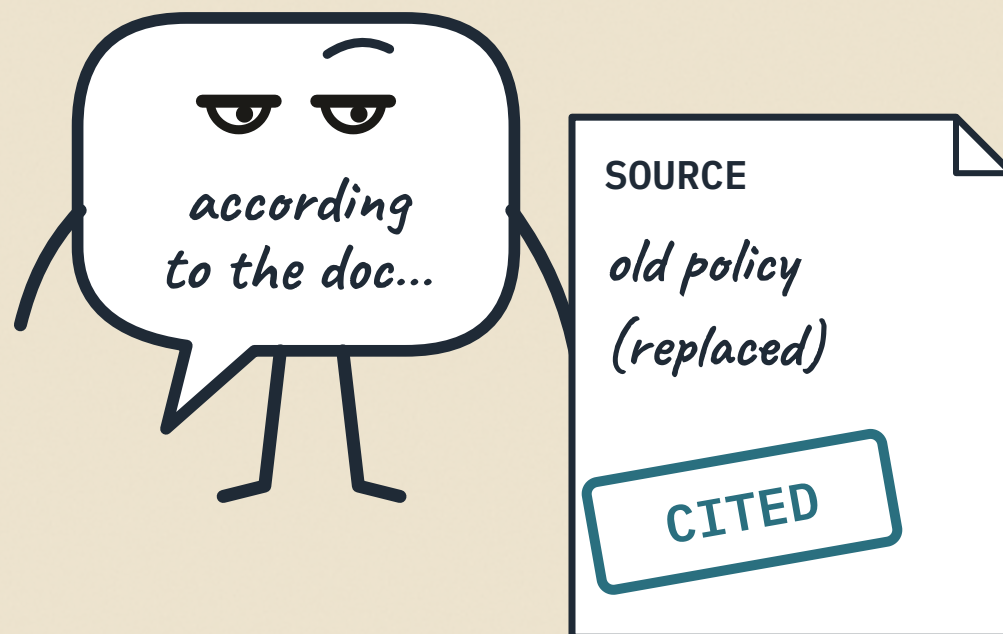
Before blaming the model



Five suspects. It's usually the first four.

When a grounded answer is wrong, walk the stages: never indexed, not returned, ranked below the cut, dropped in assembly, or not used. Only the last is a model problem.

Wrong, now with a citation.



retrieval moved the problem. it didn't fix it.

Retrieval can give the model evidence. It doesn't guarantee the evidence is relevant, current or correct, or that the model uses it well. So first ask: did the evidence reach it at all?

Read the whole explanation

[pruningmypothos.com/systems/
retrieval-augmented-generation-in-plain-terms/](https://pruningmypothos.com/systems/retrieval-augmented-generation-in-plain-terms/)

